

# Skills Summary

## Quadratics: Choosing a Method and Reading the Roots

*Use this reference while completing your worksheet or when studying.*

**Decide before you solve.** Put the equation in the form  $ax^2 + bx + c = 0$  (or leave it as a perfect square), then:

- no  $x$  term, or the variable sits inside one square  $\Rightarrow$  **square roots**
- factors over the integers (check a common factor first)  $\Rightarrow$  **factoring**
- anything else, or you cannot see it quickly  $\Rightarrow$  **quadratic formula** — it always works

Discriminant test:  $\Delta = b^2 - 4ac$ . A perfect-square  $\Delta$  means it factors;  $\Delta > 0$  two real roots,  $\Delta = 0$  one,  $\Delta < 0$  none.

### 1. Square roots (square-root-method)

$(x - h)^2 = k \Rightarrow x - h = \pm\sqrt{k} \Rightarrow x = h \pm \sqrt{k}$ . Always write **both** signs. Divide off the outside coefficient first.

**Example:** Solve  $(x - 1)^2 = 25$ .

**Step 1.** The variable appears only inside one square and there is no separate  $x$  term, so square roots is the shortest route.

**Step 2.**  $x - 1 = \pm 5$ , so  $x = 1 + 5$  or  $x = 1 - 5$ :  **$x = 6$  or  $x = -4$**

**Try it:** Solve  $2(x + 4)^2 = 50$ .

check:  $x = 1$  or  $x = -6$

### 2. Factoring (factoring-method)

Move everything to one side, factor, then use the **zero-product property**: if  $AB = 0$  then  $A = 0$  or  $B = 0$ .

For  $x^2 + bx + c$ : find two numbers with product  $c$  and sum  $b$ . Never divide both sides by  $x$  — that deletes the root  $x = 0$ .

**Example:** Solve  $x^2 + 2x - 15 = 0$ .

**Step 1.** It is already equal to zero and monic with small numbers, so hunt for a factor pair rather than reaching for the formula.

**Step 2.** Product  $-15$ , sum  $2$ : use  $5$  and  $-3$ , so  $(x + 5)(x - 3) = 0$  and  **$x = 3$  or  $x = -5$**

**Try it:** Solve  $x^2 - 7x + 12 = 0$ .

check:  $x = 4$  or  $x = 3$

### 3. Quadratic formula (quadratic-formula)

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Works on every quadratic. Use it the moment the factor hunt stalls — and simplify the radical instead of rounding, unless the problem asks for a decimal.

**Example:** Solve  $x^2 - 6x + 4 = 0$ .

**Step 1.** No integer pair multiplies to 4 and adds to  $-6$ , so factoring is a dead end and the formula is the efficient choice.

**Step 2.**  $\Delta = 36 - 16 = 20$ , so  $x = \frac{6 \pm \sqrt{20}}{2} = \frac{6 \pm 2\sqrt{5}}{2}$ , giving  $x = 3 + \sqrt{5}$  or  $x = 3 - \sqrt{5}$

**Try it:** Solve  $x^2 + 2x - 4 = 0$ .

check:  $x = -1 \pm \sqrt{5}$  so  $x^2 + 2x - 4 = 0$

### 4. Reading roots in context (interpret-roots-in-context)

Solve the equation first, then ask what each root *means*. A root that solves the equation can still be impossible in the story — a negative length, a negative time, a fraction of a person. Say which root answers the question and why the other is rejected, and give the answer with its unit.

**Example:** A rectangle is 3 m longer than it is wide and has area 40 square metres. Find its width  $w$ .

**Step 1.** Area =  $w(w + 3) = 40$  becomes  $w^2 + 3w - 40 = 0$ ; the numbers are small, so factor, then test each root against the picture.

**Step 2.**  $(w + 8)(w - 5) = 0$  gives  $w = 5$  or  $w = -8$ ; a width cannot be negative, so the rectangle is 5 m wide

**Try it:** A ball thrown up from the ground has height  $h = -16t^2 + 64t$  feet after  $t$  seconds. How long is it in the air?

check:  $t = 0, 4$  seconds